

The Effect of Small-World Network Topology on Algorithmic Polarisation Thresholds in a Bounded Confidence Model

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Research Cluster

Proposed Cluster: Data Science and Analytics

This project is fundamentally a network science study which is a listed sub-field of this cluster. The core methodology is processes on structured graphs and analysing how graph topology affects outcomes, placing it within the cluster’s scope.

Keywords: Opinion Dynamics, Algorithmic Bias, Bounded Confidence, Small-World Networks, Polarisation

National Interest Statement

Social media recommendation algorithms determine what news and opinions Australians see online. By preferentially showing content aligned with a user’s existing views, these systems can trap people in echo chambers and accelerate social division.

This research asks: *at what level of algorithmic aggressiveness does opinion polarization become unavoidable, and how does the structure of social relationships change that point?*

Answering this question is directly relevant to Australia, where the ACCC is currently debating how to regulate algorithmic harms. Evidence on where such tipping points lie could help policymakers design meaningful platform obligations and support a healthier Internet.

Introduction

Algorithm-driven social media platforms influence public opinion through the selective exposure of users to content similar to their existing beliefs (Perra & Rocha, 2019). This reduces the cross-opinion interactions that classical opinion dynamics models rely on to reach consensus (Deffuant, 2000). Understanding when such filtering produces qualitative shifts in collective opinion is an important question in computational social science (Castellano, 2009).

This project lies primarily within complex-systems simulation, with secondary relevance to network science and computational social science. It investigates how social network structure affects the algorithmic bias threshold at which a bounded confidence model transitions from consensus to polarization.

The study uses agent-based modelling with the Deffuant–Weisbuch bounded confidence model (BCM), where agents hold continuous opinions and update them through interactions with neighbours whose opinions fall within a confidence bound.

Research Question:

Does Watts–Strogatz network topology, specifically its independently tunable clustering coefficient, shift the critical algorithmic bias threshold γ^* at which a bounded confidence model transitions from consensus to polarization?

Research Problem

The relationship between algorithmic content filtering and opinion polarisation is widely debated, and research has converged on a family of related models.

Sîrbu et al. (2019) introduced a tunable bias parameter γ into the mean-field BCM, showing that higher γ increases fragmentation — but the model assumes a fully connected population and ignores network structure. Pansanella et al. (2022a) extended this to structured networks, comparing scale-free and random topologies, finding that topology affects fragmentation and convergence time.

However, neither scale-free (Barabási–Albert) nor random (Erdős–Rényi) networks isolate the effect of clustering. The Watts–Strogatz model controls clustering via rewiring parameter p while maintaining homogeneous degree, but has not been applied in this model family.

This project embeds the Sîrbu BCM onto Watts–Strogatz networks and compares polarisation thresholds against established BA and ER baselines.

Research Design

Model Specification

$N = 1000$ agents hold continuous opinions $x_i \in [0, 1]$. At each timestep agents activate once in random order. Agent i selects neighbour j with probability

$$P(j|i) \propto \exp(-\gamma|x_i - x_j|)$$

where $\gamma \geq 0$ is algorithmic bias strength. If $|x_i - x_j| < \epsilon$, opinions update via the Deffuant rule

$$x_i(t+1) = x_i(t) + \mu[x_j(t) - x_i(t)], \quad \mu = 0.5$$

otherwise no update occurs.

Network Topologies

- **Watts–Strogatz (WS):** Ring lattice, $k = 6$, rewiring $p \in \{0, 0.01, 0.05, 0.1, 0.5, 1\}$.

- **Barabási–Albert (BA)**: Scale-free, $m = 3$.
- **Erdős–Rényi (ER)**: Random graph, $\langle k \rangle \approx 6$.

Parameter Sweep

Parameter	Values
Algorithmic bias (γ)	0, 0.5, 1, 1.5, 2, 2.5, 3, 4, 5
Confidence bound (ϵ)	0.1, 0.2, 0.3, 0.4, 0.5
Network topology	WS (p values), BA, ER
Initial distribution	Uniform; Bimodal (peaks at 0.2, 0.8)
Runs per condition	50
Total	$720 \times 50 = 36,000$ simulations

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Validation

At $\gamma = 0$, results on ER and BA must reproduce the consensus boundary $\epsilon^* \approx 0.3$ from Deffuant et al. before novel parameter space is explored.

Implementation

Simulations are implemented in Python using NetworkX for graph generation and NumPy for the agent update loop.

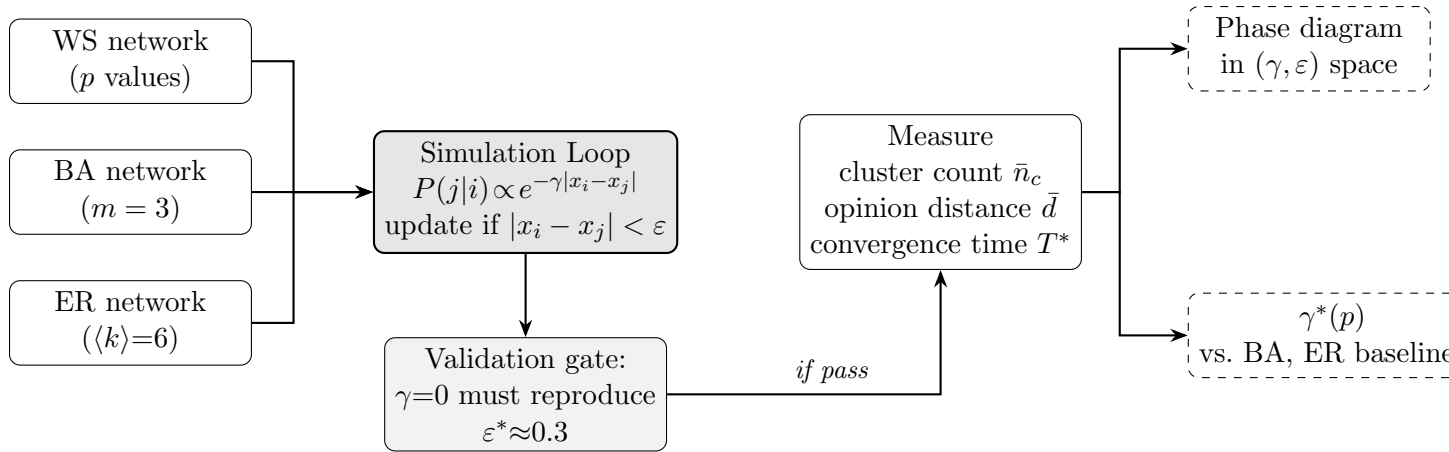


Figure 1: Methodology pipeline. Three network families feed into the simulation loop. A validation gate at $\gamma = 0$ must be passed before novel results are interpreted. Outputs are a phase diagram and topology-dependent threshold estimates $\gamma^*(p)$.

Evaluation Criteria

Model validity. At $\gamma = 0$, ER and BA networks must reproduce $\epsilon^* \approx 0.3$. Necessary because any deviation indicates an implementation error.

Baseline replication. BA and ER results with $\gamma > 0$ must match the qualitative trends of Pansanella et al. (2022a). Confirms model behaves consistently with established work.

Novel WS results. Mean cluster count is plotted against γ for each (p, ϵ) configuration, and $\gamma^*(p)$ is estimated across 50 runs. If highly clustered networks ($p \rightarrow 0$) yield lower γ^* , this supports the hypothesis that clustering accelerates polarisation.

Ethics

This study involves no human participants or personal data. All experiments are conducted on simulated agent populations.

Conclusion

This proposal investigates whether clustering in Watts–Strogatz networks shifts the polarization threshold in algorithmic bounded confidence models. Comparing results with scale-free and random baselines will determine whether clustering independently influences the emergence of polarization.

Reflection

I used Generative AI to help understand the papers that later became the basis for this research proposal, and to help clarify my writing to be more precise. Generative AI’s biggest advantage was its ability to quickly explain concepts and processes described by the papers. However, it failed to describe a research gap, and would constantly reference already existing papers as “novel ideas”. I also used Generative AI writing detection tools to assess my own writing, but they are extremely biased against technical prose. Even quoting directly from the papers themselves results in complaints of “technical jargon” and writing being too “mechanically precise”.

Annotated Bibliography

Deffuant, G., Neau, D., Amblard, F., & Weisbuch, G. (2000). Mixing beliefs among interacting agents. *Advances in Complex Systems*, 3(1–4), 87–98.

The foundational paper for the opinion model used in this project. Agents only update their opinions when they are already close to each other, and ignore those they strongly disagree with. Establishes that consensus emerges when the tolerance threshold exceeds $\epsilon^* \approx 0.3$.

Sîrbu, A., Pedreschi, D., Giannotti, F., & Kertész, J. (2019). Algorithmic bias amplifies opinion fragmentation and polarization: A bounded confidence model. *PLOS*

ONE, 14(3), e0213246.

Adds an algorithmic bias parameter γ to the model: higher values make agents more likely to interact with people they already agree with, mimicking social media recommendations. Shows this increases polarisation and slows consensus. Assumes everyone can talk to everyone, ignoring real network structure.

Perra, N., & Rocha, L.E.C. (2019). Modelling opinion dynamics in the age of algorithmic personalisation. *Scientific Reports*, 9, 7261.

Tests algorithmic filtering across several network types including Watts–Strogatz, finding that local clustering, not the presence of hubs, is what drives echo chamber formation. Uses binary rather than continuous opinions. Provides the key theoretical prediction this project tests: that more clustered networks should polarise at lower levels of algorithmic bias.

Pansanella, V., Rossetti, G., & Milli, L. (2022a). From mean-field to complex topologies: Network effects on the algorithmic bias model. *Complex Networks & Their Applications X*, Springer LNCS, pp. 329–340.

Takes the Sirbu model and runs it on scale-free and random networks, showing that network structure affects how quickly and severely opinions fragment. The closest existing work to this project. Watts–Strogatz networks are never tested, which is the gap this project fills. Also provides the baseline results this project must replicate.

Castellano, C., Fortunato, S., & Loreto, V. (2009). Statistical physics of social dynamics. *Reviews of Modern Physics*, 81, 591.

A broad review showing that opinion dynamics can be understood using the same mathematical tools physicists use to study phase transitions (sudden shifts between distinct states). This framing shapes the research question: rather than asking whether polarisation increases gradually, this project asks whether there is a sharp tipping point as algorithmic bias increases.